Technion Israel Institute of Technology

High Speed Digital Systems Lab
Electrical Engineering Department

# Portable Electronic Toolbox

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# Project Goal

A handy portable tool to an engineer

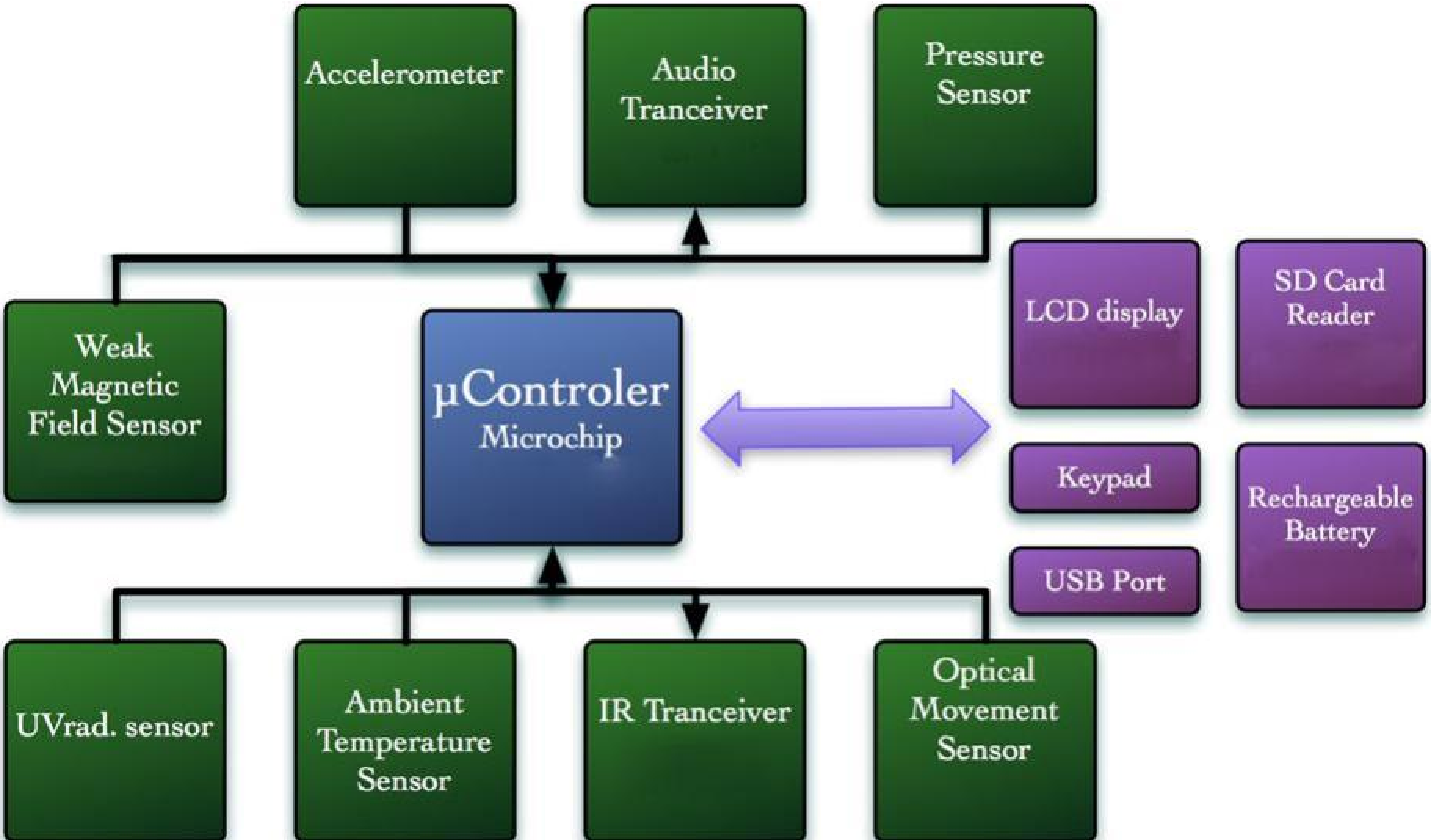
That can sense its environment

And do diverse functions

# PET Features

- Portable & user friendly
- Low-priced
- Versatile & multi-purposed
- Measure and display results instantly
- Store data for later re-evaluation
- Control over environment
- Customization and extension of feature-set in HW & SW

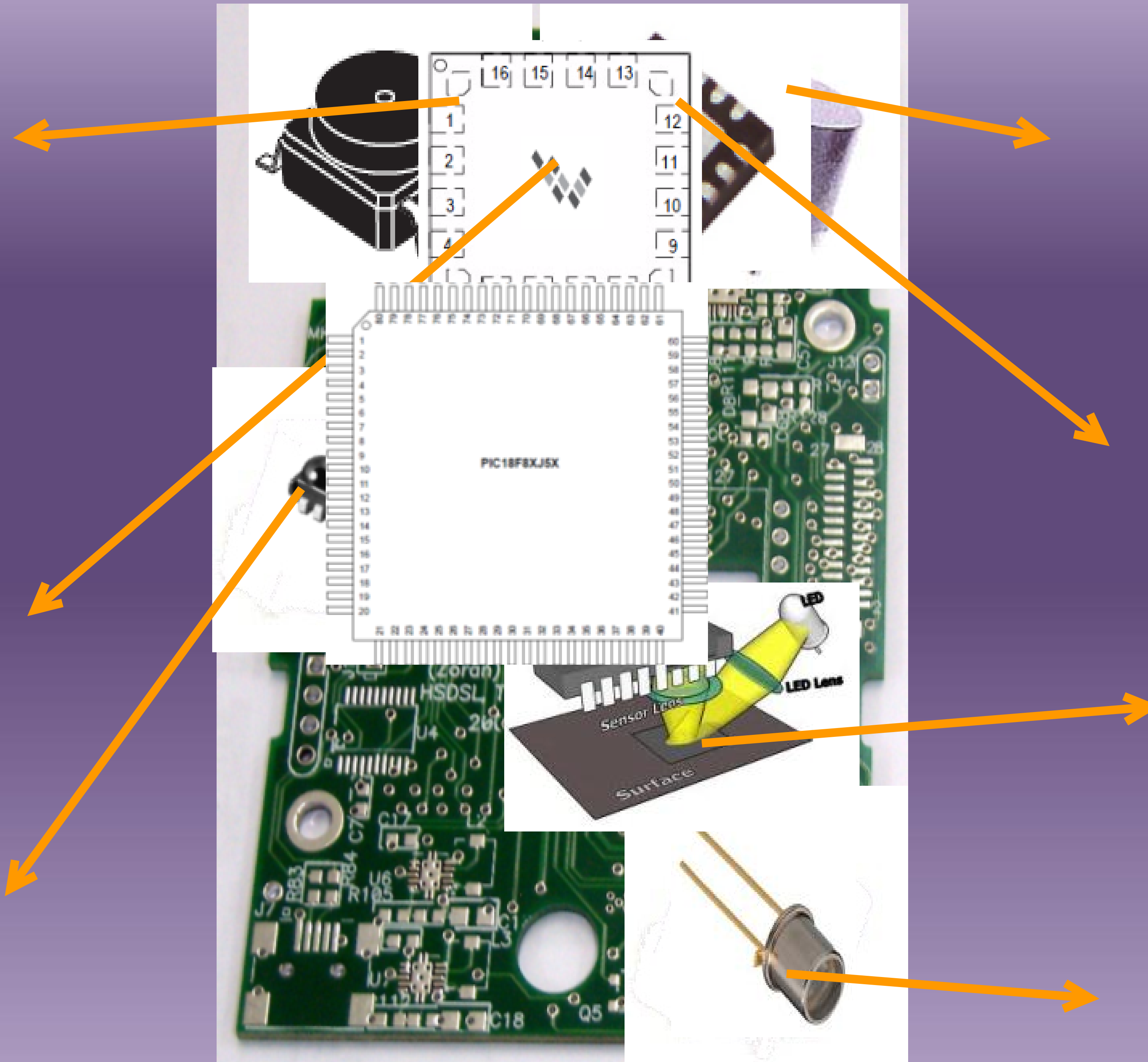
# System Block Diagram



# Development Flow

- Mechanical design in a 6.1x10x2.5cm box + 3D model
- Elec
- Layo
- Man
- Hard
- Man


- Software development environment


PIC18F8XJ5X
LED
LED Lens
Surface

# Software Development Environment

Applications

Software #1 Software #2

Operating system

Main Software

Drivers

Sensors, LCD, keyboard

Hardware

Receive analogic signals

PCD Voltage is: 3.524

1. Accelerometer
2. Pressure Sensor (kPa)
3. Magnetic Sensor
4. UV Sensor
6. Temperature – Shuts DC
7. Buzzer Testings

DC Power supply is OFF

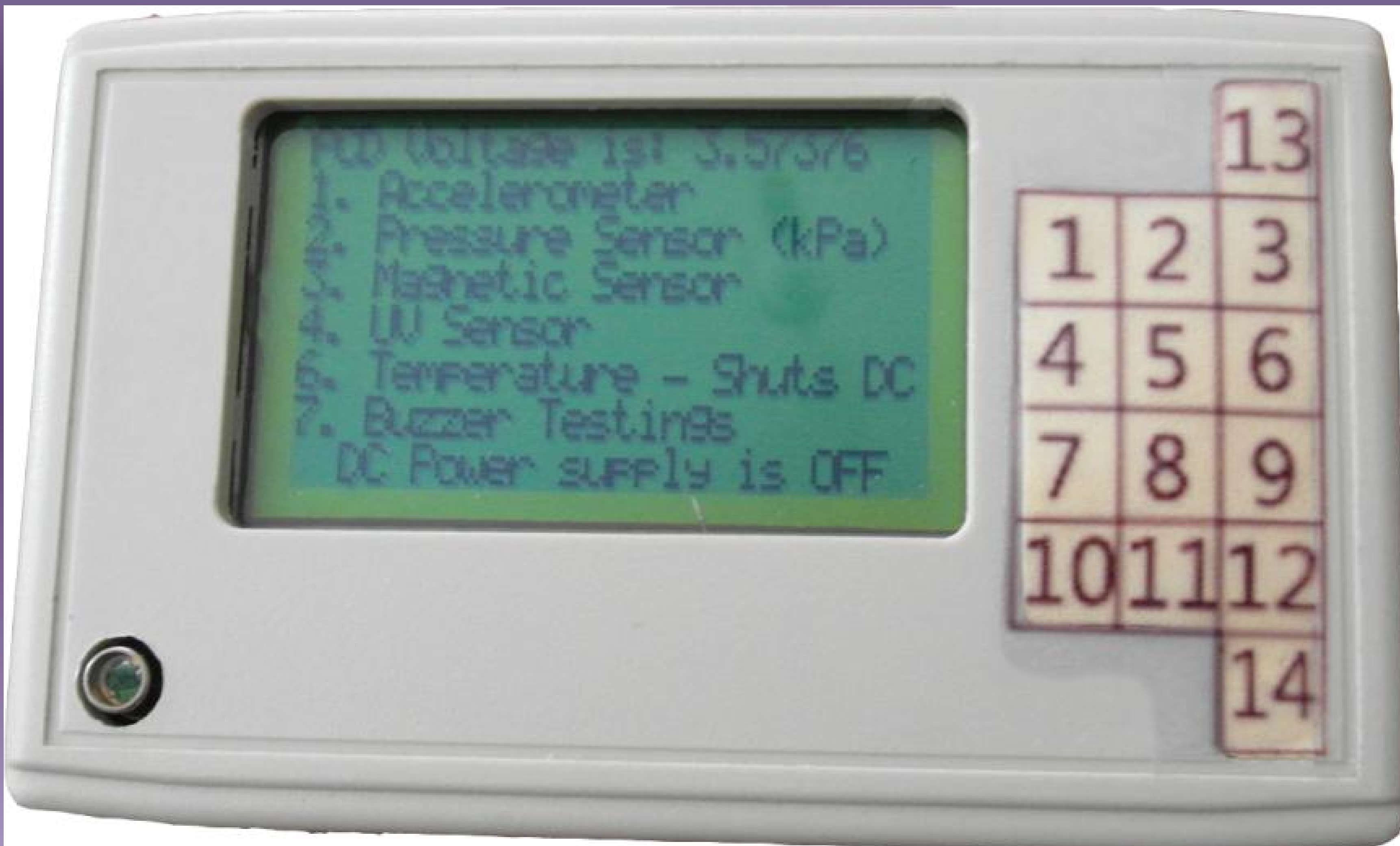
# Future Applications

- GUI for PC
- IR - remote control
- IR + microphone – volume control
- Optic – digital ruler
- Optic – OCR
- Accelerometer – digital leveler
- Accelerometer – angle gauge
- UV + buzzer – high radiation indicator

# Our personal gains

- Handling a variety of different issues and integrating them.
- Approaching a problem from various sides and knowing where to look for quick and efficient solutions.
- Accompanying a project from an idea to final complete product.

# The PET

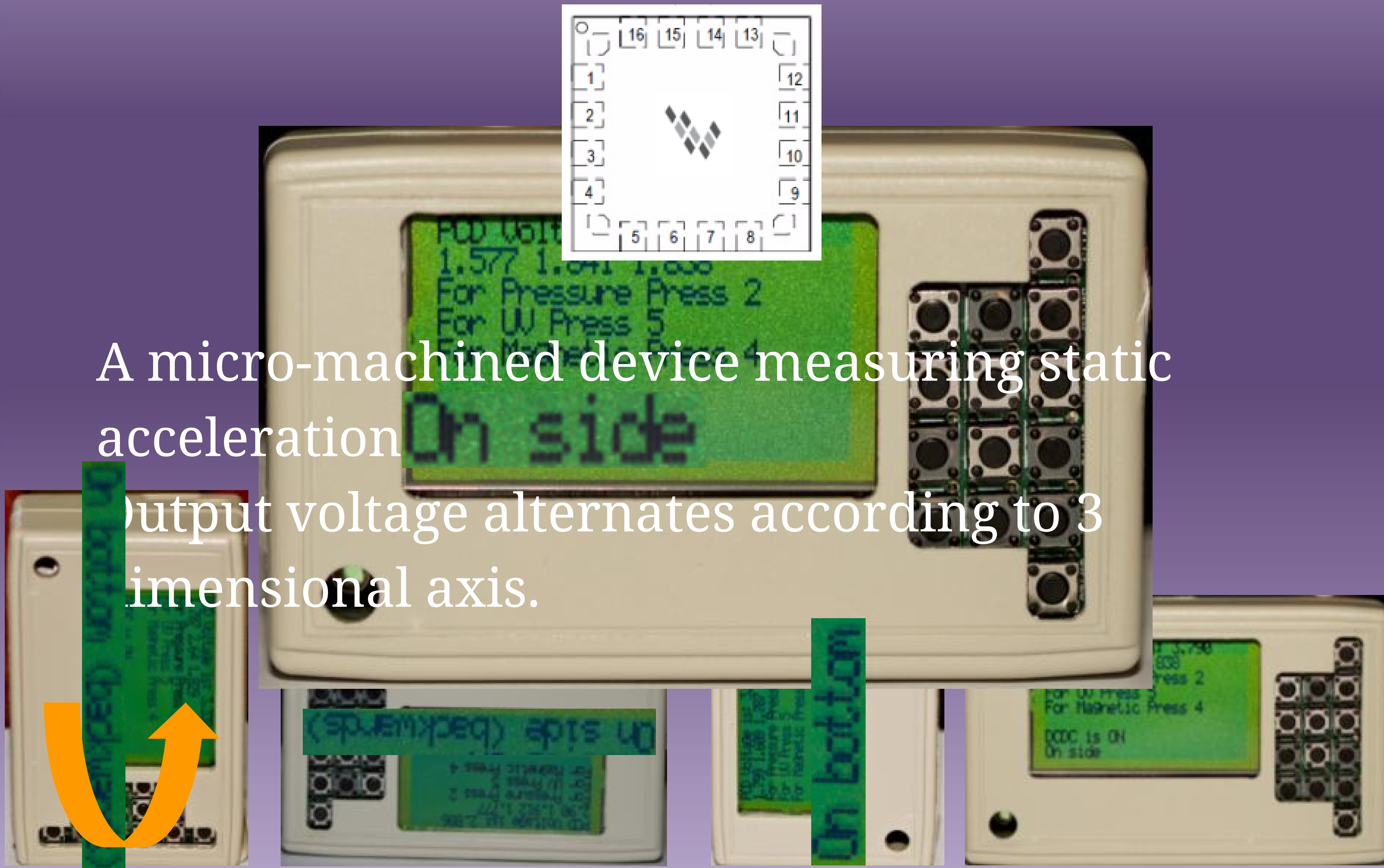

1. Accelerometer
2. Pressure Sensor (kPa)
3. Magnetic Sensor
4. UV Sensor
6. Temperature - Shuts DC
7. Buzzer Testings
DC Power supply is OFF
13
1 2 3
4 5 6
7 8 9
10 11 12
14

# Try It Yourself!

- PET #1 – motion detector
  - Press red button to open and play!
- PET #2 – orientation detector
  - Press red button to open. Press number 1 on keyboard and change PETs' orientation
- PET #3 – temperature
  - Press red button to open. Press number 6 on keyboard


THANK
YOU
THANK

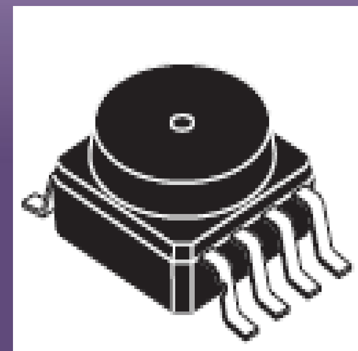
# Accelerometer

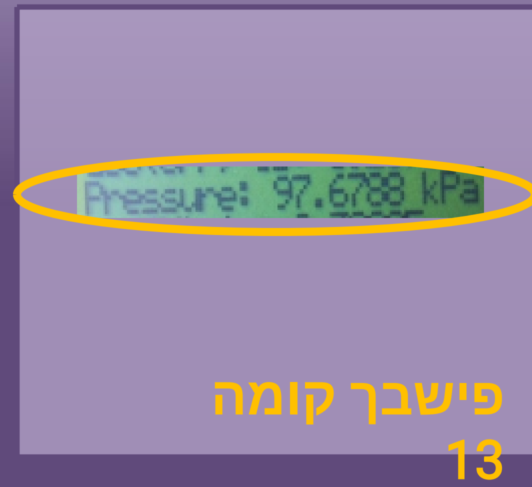



A micro-machined device measuring static acceleration

Output voltage alternates according to 3 dimensional axis.

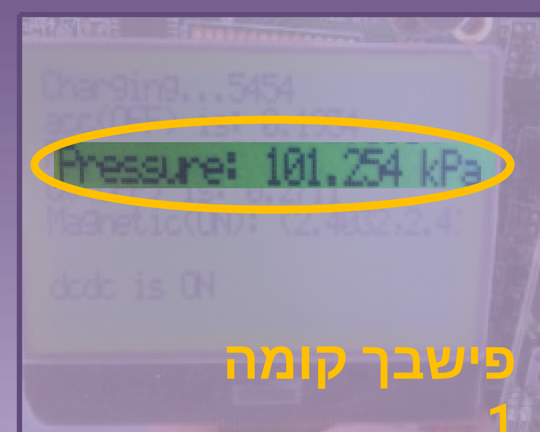
# Pressure






A silicon sensor measuring absolute pressure.
Output voltage can be converted to

pressure by: $$P = \frac{Vout + 0.475}{0.045}$$



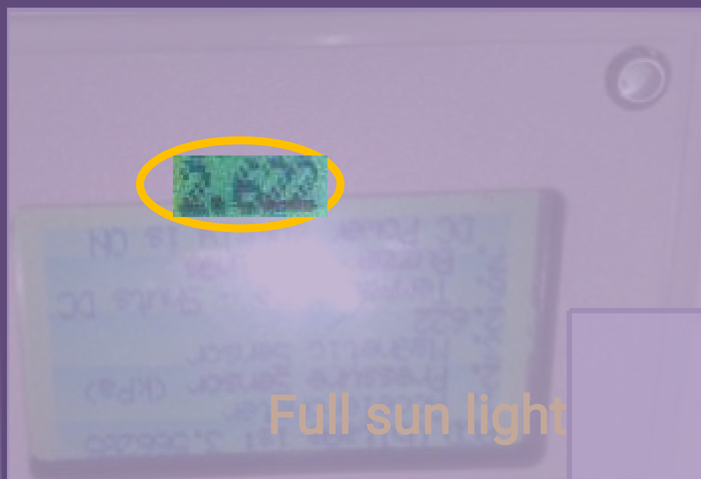
# UV Sensor

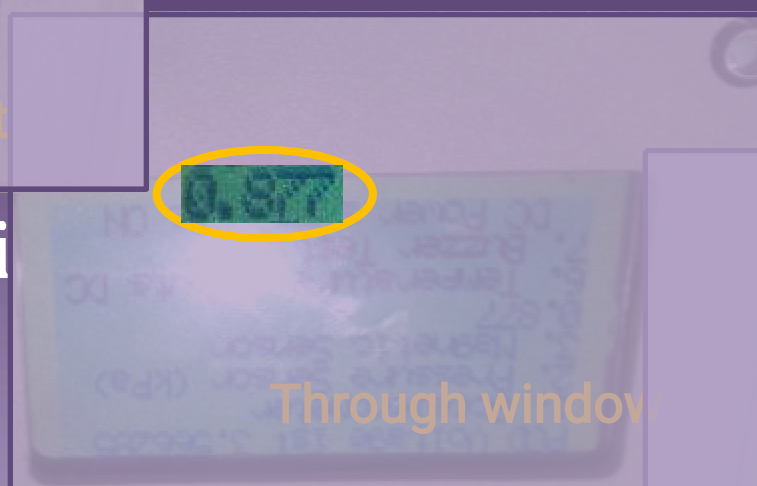
otodiode for UV sensing.

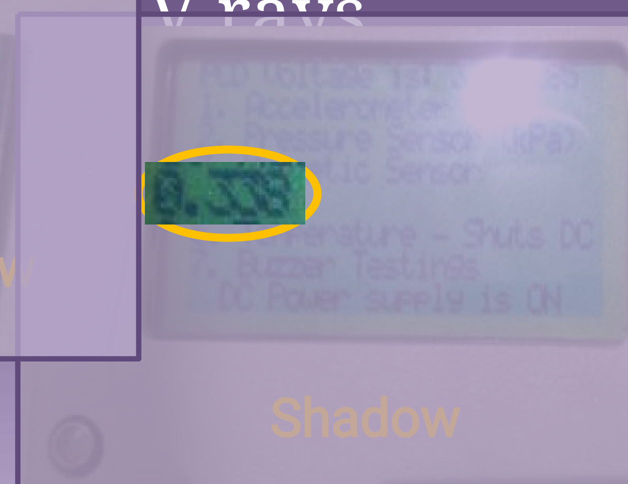
ce of the box

V rays

Max value i



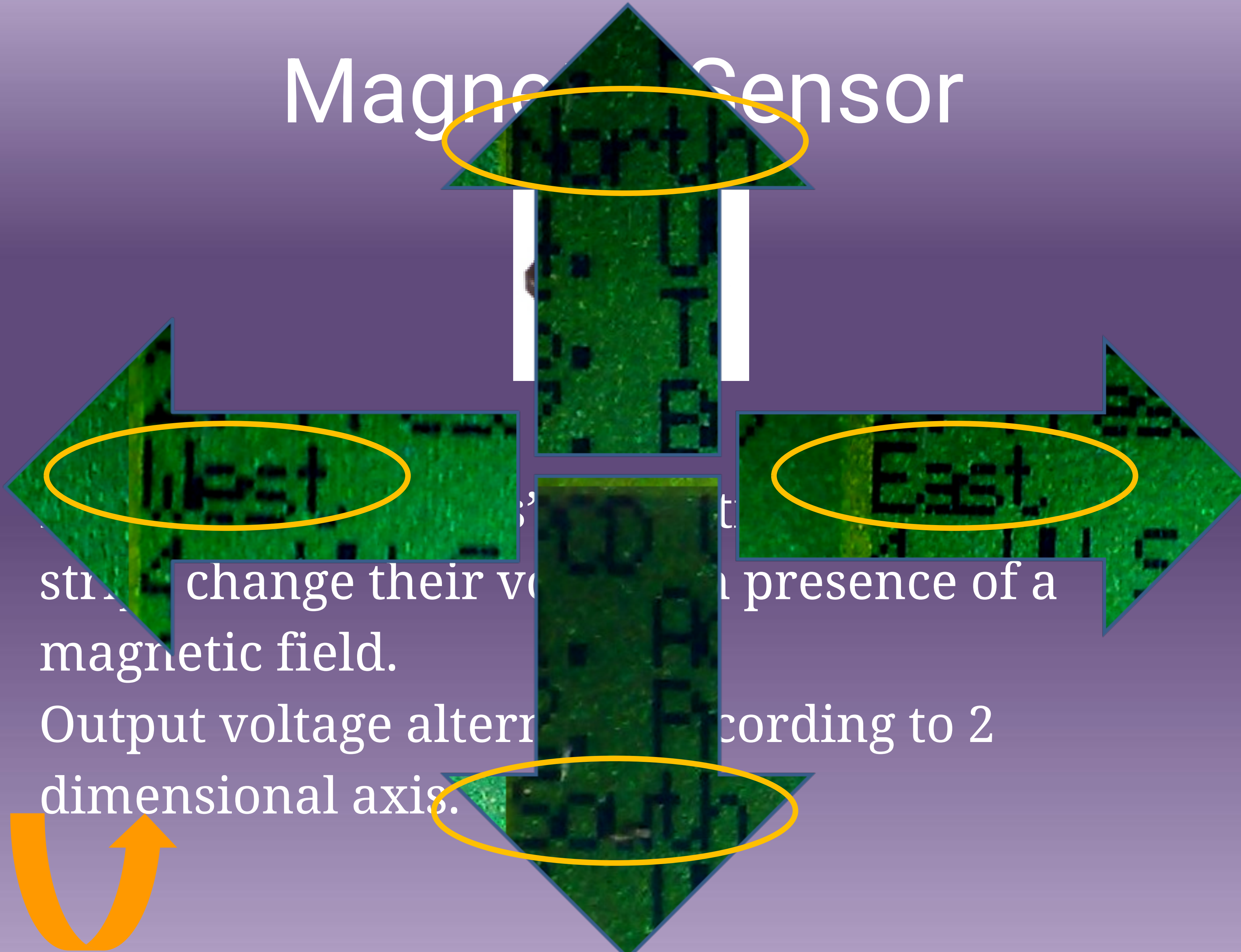



# IR

IR transmitter/receiver.
Receiving function is checked using ISR due to high frequencies.

# Magne Sensor

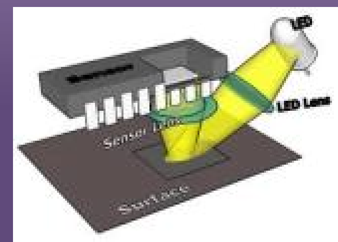



s' t

str change their vo presence of a magnetic field.

Output voltage altern cording to 2 dimensional axis.

# Optic Sensor




A CMOS optic sensor that decides on direction of
mover … ond and
compa … ernal
led an …
Outpu … essed in
a state … on.